

CHAPTER C8

RAIN LOADS

C8.1 DEFINITIONS AND SYMBOLS

C8.1.2 Symbols

- A = Tributary roof area, plus one-half the wall area that diverts rainwater onto the roof, serviced by a single drain outlet in the secondary drainage system, in ft^2 (m^2).
- D = Drain bowl diameter for a primary roof drain, or overflow dam or standpipe diameter for a secondary roof drain, in in. (mm).
- i = Design rainfall intensity, in./h (mm/h).
- Q = Flow rate out of a single drainage system, in gal./min (m^3/s).
- L_r = Length of level roof edge that allows for free overflow drainage of rainwater when the roof edge is acting as the secondary drainage system, in ft (m).

C8.2 ROOF DRAINAGE

Roof drainage systems are not always designed to handle all the flow associated with intense, short-duration rainfall events. For example, the International Plumbing Code (ICC 2012) uses a 1-h duration event with a 100-yr return period for the design of both the primary and secondary drainage systems. An adequate secondary (overflow) drainage system, which is used to limit the depth of water on the roof in the event of clogging of the primary drains, must be designed for an adequately short-duration rainfall event. Some plumbing codes use an arbitrary 1-hour duration storm event for the design of roof drain systems; however, the critical duration for a roof is generally closer to 15 min (the critical duration depends on the roof geometry and drain sizes), and therefore the plumbing codes do not appropriately account for the coincidence of both blocked primary drains and short-duration rainfall events at the design mean recurrence interval (return period or frequency). Graber (2009) provides guidance for determining the critical durations for different types of roof configurations. A very severe local storm or thunderstorm in excess of the 100-year return period storm may produce a deluge of such intensity and duration that properly designed primary drainage systems are temporarily overloaded. Such temporary loads are typically covered in design when blocked drains (see Section 8.3) and a rainfall duration of 15 min are considered. The use of a 60-min duration/100-year return period rainfall event for the design of the primary drainage system and the 15-min duration/100-year return period rainfall event for the secondary drainage system (assuming the primary drainage system is completely blocked) is consistent with the NFPA 5000 Building Construction and Safety Code (2012). Internal gutters are typically designed for 2-min to 5-min duration rainfall events since their critical duration is very short due to their limited storage volume and inability to attenuate a rainfall event.

The National Oceanic and Atmospheric Administration (NOAA's) National Weather Service Precipitation Frequency Data Server, Hydrometeorological Design Studies Center provides rainfall intensity data in inches per hour for the 15-min duration/100-year mean recurrence interval (<http://hdsc.nws.noaa.gov/hdsc/pfds/index.html>). Precipitation intensity (i in Eq. [C8.3-1]) is in the units of inches per hour; if precipitation depth is provided, a conversion to intensity is required.

The following roof conditions adversely affect the critical duration, or increase the peak flow rate, and should be avoided or appropriately considered by the designer when determining the design rain load:

1. Roofs with internal gutters that have limited storage capacity and quickly fill with rainwater. Gutters are typically designed for 2- to 5-min duration storms since their critical duration is much shorter than the critical duration for typical roofs with scuppers or internal drains.
2. Architecturally complex roofs with internal gutters with significant gutter slopes. Significant gutter slopes allow water to flow at high velocities, which need to be considered when designing the gutter outlets and determining rain loads.
3. Areas susceptible to a concentration of flow, for example, when an addition is added to a low-sloped gable roof (Fig. C8.2-1). In this case, rainwater at the edge of the main roof cannot build over the primary or secondary drains to attain the design flow rate, and the water flows into and inundates the small roof extension.
4. Small roofs adjacent to large walls, where the wall is capable of contributing substantial wind-driven rain flow (sheet flow down the wall) to the roof.
5. High roof areas that drain onto a low roof, increasing the tributary roof area and decreasing the critical rainfall duration.

Item Nos. 4 and 5 can occur when building additions occur, i.e., the new construction imposes an unfavorable condition on the existing construction conditions.

Roof drainage is a structural, architectural, and mechanical (plumbing) issue. The type and location of secondary drains and the hydraulic head above their inlets at the design flow must be known in order to determine rain loads. Design team coordination is particularly important when establishing rain loads.

C8.3 DESIGN RAIN LOADS

The amount of water that could accumulate on a roof from blockage of the primary drainage system is determined, and the roof is designed to withstand the load created by that water plus the uniform load caused by water that rises above the inlet of the

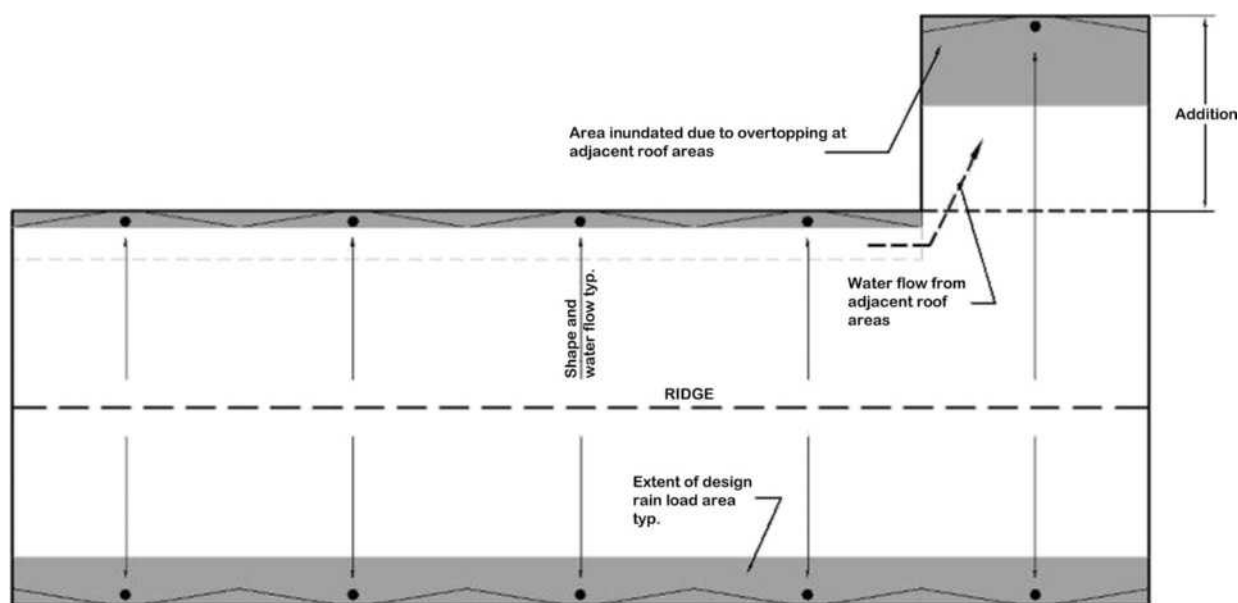


FIGURE C8.2-1 Low-Slope Gable Roof with Drainage Condition at Building Extension

secondary drainage systems at its design flow. If parapet walls, cant strips, expansion joints, and other features create the potential for deep water in an area, it may be advisable to install in that area secondary (overflow) drains with separate drain lines rather than overflow scuppers to reduce the magnitude of the design rain load. Where geometry permits, free discharge is the preferred form of emergency drainage.

When determining these water loads, it is assumed that the roof does not deflect. This assumption eliminates complexities associated with determining the distribution of water loads within deflection depressions. However, it is quite important to consider this water when assessing ponding instability in Section 8.4.

The depth of water, d_h , above the inlet of the secondary drainage system (i.e., the hydraulic head) is a function of the rainfall intensity, i , at the site, the area of roof serviced by that drainage system, and the size of the drainage system.

The flow rate through a single drainage system is as follows:

$$Q = 0.0104Ai \quad (\text{C8.3-1})$$

$$Q = 0.278 \times 10^{-6}Ai \quad (\text{C8.3-1.si})$$

The hydraulic head, d_h , is related to flow rate, Q , for various drainage systems in Table C8.3-1. This table indicates that d_h can vary considerably depending on the type and size of each drainage system and the flow rate it must handle. For this reason, the single value of 1 in. (25 mm) (i.e., 5 lb/ft² [0.24 kN/m²]) used in ASCE 7-93 was eliminated.

The hydraulic head, d_h , can generally be assumed to be negligible for design purposes when the secondary drainage system is free to overflow along a roof edge where the length of the level roof edge (L_r) providing free drainage is

$$L_r \geq Ai/400 \quad (\text{C8.3-2})$$

$$L_r \geq Ai/3,100 \quad (\text{C8.3-2.si})$$

Eq. (C8.3-2) is based on the assumption that hydraulic head (d_h) of approximately 0.25 in. (6 mm) above the level roof edge,

which represents a rain load of 1.3 lb/ft² (6.3 kg/m²), is negligible in most circumstances.

Flow rates and corresponding hydraulic heads for roof drains are often not available in industry codes, standards, or drain manufacturers' literature for many commonly specified drain types and sizes. Since the hydraulic characteristics and performance of roof drains can depend not only on the size of the drain outlet but also on the geometry of the drain body (e.g., the diameter of the drain dam and depth of the drain bowl), determining the flow rate and corresponding hydraulic head for a drain can be difficult based only on hydraulic calculations. This is particularly true when considering the difficulty in predicting the flow regime (i.e., weir flow, orifice flow, or transition between the two) and the significant effect that flow regime has on the relationship between flow rate and corresponding hydraulic head for a drain.

Based on a drain flow testing program completed by FM Global (2012), the hydraulic heads corresponding to a given range of drain flow rates are provided in Tables C8.3-1 and C8.3-2. This drain testing program included six sizes of primary roof drains and seven sizes and types of secondary (overflow) roof drains. The drains were tested with debris guards (strainers) in place and in a test basin with a relatively smooth bottom surface (waterproofing membrane) to simulate typical smooth-surface roofing material. Measurements of water depth in the test basin were made at a distance of 2 ft (0.6 m) or more from the drain, which ensured that the head measurements were not significantly affected by surface water velocity, and therefore were made where the velocity head was negligible, which was confirmed when comparing water depth based on direct depth measurements to hydraulic head based on pressure taps embedded in the bottom surface of the test basin.

Refer to Fig. C8.3-1 for a schematic view of a secondary drain and the relationship between the drain, the roof surface, and the head.

The following method can be used to approximate hydraulic head for differing drain body dimensions:

(a) For weir flow and transition flow regime designations (cells that are not shaded) in Tables C8.3-1 and C8.3-2:

Table C8.3-1 Flow Rate (Q) in Gallons per Minute for Secondary (Overflow) Roof Drains at Various Hydraulic Heads (d_h) above the Dam or Standpipe, in Inches

Flow rate (gal./min)	Hydraulic Head (in.) above Dam or Standpipe						
	Overflow Dam 8 in. Diameter			Overflow Dam 12.75 in. Diameter	Overflow Dam 17 in. Diameter	Overflow Standpipe 6 in. Diameter	
	Drain Outlet Size (in.)						
	3	4	6	6	8	10	4
	Drain Bowl Depth (in.)						
	2	2	2	2	3.25	4.25	2
50	0.5	0.5	0.5	0.5	0.5	—	1.0
75	1.0	—	—	—	—	—	—
100	1.5	1.0	1.0	1.0	0.5	1.0	1.5
125	2.0	—	—	—	—	—	—
150	2.0	1.5	1.5	1.0	—	—	2.5
175	3.0	—	—	—	—	—	—
200	—	2.0	2.0	1.5	1.5	1.5	2.5
225	—	—	—	—	—	—	—
250	—	2.5	2.5	1.5	—	—	2.5
300	—	3.0	3.0	2.0	2.0	1.5	3.0
350	—	3.5	3.5	2.5	—	—	3.5
400	—	5.5	3.5	3.0	2.5	2.0	—
450	—	—	4.0	3.0	—	—	—
500	—	—	5.0	3.5	3.0	2.5	—
550	—	—	5.5	4.0	—	—	—
600	—	—	6.0	5.5	3.5	2.5	—
650	—	—	—	—	—	—	—
700	—	—	—	—	3.5	3.0	—
800	—	—	—	—	4.5	3.0	—
900	—	—	—	—	5.0	3.5	—
1,000	—	—	—	—	5.5	3.5	—
1,100	—	—	—	—	—	4.0	—
1,200	—	—	—	—	—	4.5	—

Notes:

1. Assume that the flow regime is either weir flow or transition flow, except where the hydraulic head values are in shaded cells below the heavy line that designates orifice flow.
2. To determine total head, add the depth of water (static head, d_s) above the roof surface to the secondary drain inlet (which is the height of the dam or standpipe above the roof surface) to the hydraulic head listed in this table.
3. Linear interpolation for flow rate and hydraulic head is appropriate for approximations.
4. Extrapolation is not appropriate.

Source: Adapted from FM Global (2012).

Where the specified secondary (overflow) drain dam or standpipe diameter differs from what is provided in Tables C8.3-1 and C8.3-2, the hydraulic head can be adjusted based on Eq. (C8.3-3) while holding flow rate constant; however, it is advisable not to use an adjusted design hydraulic head less than 80% of the hydraulic head indicated in the tables (for a given flow rate) unless flow test results are provided to justify the hydraulic head values.

$$d_{h2} = [(D_1/D_2)^{0.67}](d_{h1}) \quad (\text{C8.3-3})$$

where

d_{h1} = known hydraulic head from Tables C8.3-1 and C8.3-2.

D_1 = overflow dam or standpipe diameter for secondary (overflow) drain, corresponding to d_{h1} for a given flow rate, as shown in Tables C8.3-1 and C8.3-2.

d_{h2} = hydraulic head to be determined for the specified secondary drain.

D_2 = specified overflow dam or standpipe diameter for secondary (overflow) drain corresponding to d_{h2} for a given flow rate.

Example 1: Determine the total head for an 8-in. secondary drain (8-in. outlet diameter) with a 10-in. diameter $\times$ 2 in. high overflow dam (d_s) at a flow rate (Q) of 300 gal./min.

From Table C8.3-1:

$D_1 = 12.75$ in. (dam diameter).

$d_{h1} = 2.0$ in. for 300 gal./min, 8-in. outlet.

For the specified 10-in. diameter overflow dam on an 8-in. drain outlet:

$D_2 = 10$ in. (dam diameter).

Therefore,

$$d_{h2} = [(D_1/D_2)^{0.67}](d_{h1})$$

$$d_{h2} = [(12.75 \text{ in.}/10 \text{ in.})^{0.67}](2.0 \text{ in.}) = 2.4 \text{ in. at } Q = 300 \text{ gal./min}$$

$$\text{Total head} = d_{h2} + d_s = 2.4 \text{ in.} + 2 \text{ in.} = 4.4 \text{ in.}$$

(b) For orifice flow regime designations for roof drains, as shown in the shaded cells in Tables C8.3-1 and C8.3-2:

Table C8.3-2 Flow Rate (Q) in Cubic Meters per Second for Secondary (Overflow) Roof Drains at Various Hydraulic Heads (d_h) above the Dam or Standpipe, in Millimeters

Flow Rate (m³/s)	Hydraulic Head (mm) above Dam or Standpipe						
	Overflow Dam 203 mm Diameter			Overflow Dam 329 mm Diameter		Overflow Dam 432 mm Diameter	Overflow Standpipe 152 mm Diameter
	Drain Outlet Size (mm)						
	76	102	152	152	203	254	102
	Drain Bowl Depth (mm)						
	51	51	51	51	83	108	51
0.0032	13	13	13	13	13	—	25
0.0047	25	—	—	—	—	—	—
0.0063	38	25	25	25	13	25	38
0.0079	51	—	—	—	—	—	—
0.0095	51	38	38	25	—	—	64
0.0110	76	—	—	—	—	—	—
0.0126	—	51	51	38	38	38	64
0.0142	—	—	—	—	—	—	—
0.0158	—	64	64	38	—	—	64
0.0189	—	76	76	51	51	38	76
0.0221	—	89	89	64	—	—	89
0.0252	—	140	89	76	64	51	—
0.0284	—	—	102	76	—	—	—
0.0315	—	—	127	89	76	64	—
0.0347	—	—	140	102	—	—	—
0.0379	—	—	152	140	89	64	—
0.0410	—	—	—	—	—	—	—
0.0442	—	—	—	—	89	76	—
0.0505	—	—	—	—	114	76	—
0.0568	—	—	—	—	127	89	—
0.0631	—	—	—	—	140	89	—
0.0694	—	—	—	—	—	102	—
0.0757	—	—	—	—	—	114	—

- Notes:
- 1. Assume that the flow regime is either weir flow or transition flow, except where the hydraulic head values are in shaded cells below the heavy line that designates orifice flow.
 - 2. To determine total head, add the depth of water (static head, d_s) above the roof surface to the secondary drain inlet (which is the height of the dam or standpipe above the roof surface) to the hydraulic head listed in this table.
 - 3. Linear interpolation for flow rate and hydraulic head is appropriate for approximations.
 - 4. Extrapolation is not appropriate.

Source: Adapted from FM Global (2012).

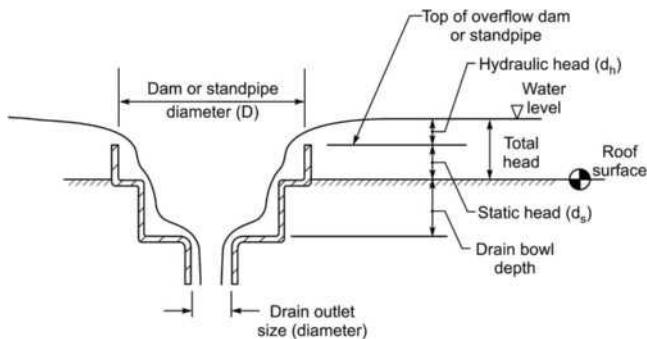


FIGURE C8.3-1 Schematic Cross Section of Secondary (Overflow) Roof Drain and Total Head ($d_s + d_h$). Drain Debris Guard (Strainer) and Ring Clamp (Gravel Stop) Not Shown for Clarity

The depth of the drain bowl can affect the hydraulic head acting on the drain outlet for a given flow rate; therefore, where the depth of the specified drain bowl is less than the depth of the tested drain bowl (indicated in the tables), the difference in drain bowl depth should be added to the hydraulic head from the tables to determine the design hydraulic head and total head. Where the depth of the specified drain bowl is greater than that indicated in the tables, the difference in drain bowl depth can be subtracted from hydraulic head in the tables to determine the design hydraulic head and total head; however, it is advisable not to use an adjusted design hydraulic head less than 80% of the hydraulic head provided in the tables (for a given flow rate) unless flow test results are provided to justify the hydraulic head values.

Example 2: Determine the total head for a 4 in. secondary drain (4-in. outlet diameter) with a drain bowl depth of 1.5 in. and a 2.5-in.-high overflow dam (d_s), at a flow rate (Q) of 350 gal./min.

Table C8.3-3 Flow Rate, Q , in Gallons Per Minute for Scuppers at Various Hydraulic Heads (d_h) in Inches

Drainage System	Hydraulic Head, d_h , in.									
	1	2	2.5	3	3.5	4	4.5	5	7	8
6-in. wide channel scupper ^a	18	50	<i>b</i>	90	<i>b</i>	140	<i>b</i>	194	321	393
24-in. wide channel scupper	72	200	<i>b</i>	360	<i>b</i>	560	<i>b</i>	776	1,284	1,572
6-in. wide, 4-in. high, closed scupper ^a	18	50	<i>b</i>	90	<i>b</i>	140	<i>b</i>	177	231	253
24-in. wide, 4-in. high, closed scupper	72	200	<i>b</i>	360	<i>b</i>	560	<i>b</i>	708	924	1,012
6-in. wide, 6-in. high, closed scupper	18	50	<i>b</i>	90	<i>b</i>	140	<i>b</i>	194	303	343
24-in. wide, 6-in. high, closed scupper	72	200	<i>b</i>	360	<i>b</i>	560	<i>b</i>	776	1,212	1,372

^aChannel scuppers are open-topped (i.e., three-sided). Closed scuppers are four-sided.

^bInterpolation is appropriate, including between widths of each scupper.

Source: Adapted from FM Global (2012).

Table C8.3-4 In SI, Flow Rate, Q , in Cubic Meters Per Second for Scuppers at Various Hydraulic Heads (d_h) in Millimeters

Drainage System	Hydraulic Head d_h , mm									
	25	51	64	76	89	102	114	127	178	203
152-mm wide channel scupper ^a	0.0011	0.0032	<i>b</i>	0.0057	<i>b</i>	0.0088	<i>b</i>	0.0122	0.0202	0.0248
610-mm wide channel scupper	0.0045	0.0126	<i>b</i>	0.0227	<i>b</i>	0.0353	<i>b</i>	0.0490	0.0810	0.0992
152-mm wide, 102-mm high, closed scupper ^a	0.0011	0.0032	<i>b</i>	0.0057	<i>b</i>	0.0088	<i>b</i>	0.0112	0.0146	0.0160
610-mm wide, 102-mm high, closed scupper	0.0045	0.0126	<i>b</i>	0.0227	<i>b</i>	0.0353	<i>b</i>	0.0447	0.0583	0.0638
152-mm wide, 152-mm high, closed scupper	0.0011	0.0032	<i>b</i>	0.0057	<i>b</i>	0.0088	<i>b</i>	0.0122	0.0191	0.0216
610-mm wide, 152-mm high, closed scupper	0.0045	0.0126	<i>b</i>	0.0227	<i>b</i>	0.0353	<i>b</i>	0.0490	0.0765	0.0866

^aChannel scuppers are open-topped (i.e., three-sided). Closed scuppers are four-sided.

^bInterpolation is appropriate, including between widths of each scupper.

Source: Adapted from FM Global (2012).

Table C8.3-5 Flow Rate (Q) in Gallons per Minute, for Circular Scuppers at Various Hydraulic Heads (d_h) in Inches

d_h (in.)	Scupper Flow (gal./min)						
	Scupper Diameter (in.)						
	5	6	8	10	12	14	16
1	6	7	8	8	10	10	10
2	25	25	30	35	40	40	45
3	50	55	65	75	75	90	95
4	80	90	110	130	140	155	160
5	115	135	165	190	220	240	260
6	155	185	230	270	300	325	360
7	190	230	300	350	410	440	480
8	220	280	375	445	510	570	610

Notes:

1. Hydraulic head (d_h) is taken above the scupper invert (design water level above base of scupper opening).
2. Linear interpolation is appropriate for approximations.
3. Extrapolation is not appropriate.

Source: Data from Carter (1957) and Bodhaine (1968).

Table C8.3-6 Flow Rate (Q) in Cubic Meters per Second for Circular Scuppers at Various Hydraulic Heads (d_h) in Millimeters

d_h (mm)	Scupper Flow Rate (m ³ /s)						
	Scupper Diameter (mm)						
	127	152	203	254	305	356	406
25	0.0004	0.0004	0.0005	0.0005	0.0006	0.0006	0.0006
51	0.0016	0.0016	0.0019	0.0022	0.0025	0.0025	0.0028
76	0.0032	0.0035	0.0041	0.0047	0.0047	0.0057	0.0060
100	0.0050	0.0057	0.0069	0.0082	0.0088	0.0098	0.0101
127	0.0073	0.0085	0.0104	0.0120	0.0139	0.0151	0.0164
152	0.0098	0.0117	0.0145	0.0170	0.0189	0.0205	0.0227
178	0.0120	0.0145	0.0189	0.0221	0.0259	0.0278	0.0303
203	0.0139	0.0177	0.0237	0.0281	0.0322	0.0360	0.0385

Notes:

1. Hydraulic head (d_h) is taken above the scupper invert (design water level above base of scupper opening).
2. Linear interpolation is appropriate for approximations.
3. Extrapolation is not appropriate.

Source: Data from Carter (1957) and Bodhaine (1968).

From Table C8.3-1: When $Q = 350$ gal./min, for a 4-in. drain with an 8-in. dam, $d_h = 3.5$ in., the flow regime is orifice flow (shaded portion of the table), and the drain bowl depth is 2 in.

The specified drain bowl depth is 1.5 in., and since this is 0.5 in. less than the drain bowl depth referenced in the table, the hydraulic head from the table is increased by 0.5 in.

Therefore, for the proposed drain: $d_h = 3.5$ in. + 0.5 in. = 4.0 in.
Total head = $d_h + d_s = 4.0$ in. + 2.5 in. = 6.5 in.

Drain outlet sizes are generally standard in the industry, so it is unlikely that adjustments to hydraulic head values in Tables C8.3-1 and C8.3-2 based on differing drain outlet sizes will be needed.

Where a roof drain is installed in a sump pan located below the adjoining roof surface, reductions in hydraulic head and rain load on the adjoining roof surface should only be credited when based on hydraulic analysis from a qualified plumbing engineer.

Refer to Tables C8.3-3, C8.3-4, C8.3-5, and C8.3-6 for flow rates of rectangular and circular (pipe) roof scuppers at various hydraulic heads. Note that these tables are based on the assumption that no backwater is present (i.e., free outfall) at the discharge end of the scupper. If backwater is present, then the hydraulic head can be expected to increase for the same flow rate.

C8.4 PONDING INSTABILITY AND PONDING LOAD

As water accumulates on roofs, roof deflection allows additional water flows to such areas, and the roof tends to deflect more, allowing a deeper pond to form there. If the structure does not possess enough stiffness to resist this progression, failure by localized overloading may result. Haussler (1962), Chinn (1965), Marino (1966), Salama and Moody (1967), Sawyer (1967, 1968), Chinn et al. (1969), Heinzerling (1971), Burgett (1973), AISC (1978), Associate Committee on the National Building Code (1990), FM Global (2012), SBCCI (1991), BOCA (1993), AISC 360 (2016), and SJI (2007) contain information on ponding and its importance in the design of flexible roofs. Rational design methods to preclude instability from ponding are presented in AISC 360 (2016) and SJI (2007). In determining ponding loads, the primary and secondary members in two-way roof framing systems are to be analyzed concurrently and iteratively since the deflection of primary members (girders) will cause deflection and increase the ponding loads on secondary members (e.g., joists, purlins, or rafters) and the deflection of secondary members will increase the deflection and ponding loads on girders. The deflection of decking can also add additional ponding loads. AISC 360 (2016), Appendix 2, provides methods to evaluate ponding stability for structural steel roof framing. Note, however, that all framing systems can be impacted by ponding.

Regardless of roof slope, if water is impounded on the roof to reach a secondary drainage system, ponding instability can occur. Where such impounded water situations exist, the bay is considered a susceptible bay. Shown in Fig. C8.4-1 are typical susceptible bays for a roof with any slope. For the same structure with secondary members perpendicular to the free draining edge and a roof slope less than 1/4 in./ft, all bays are susceptible. For the same structure with secondary members parallel to the free draining edge and a roof slope less than 1 in./ft, all bays are susceptible. Fig. C8.4-2 shows a roof with perimeter overflow (secondary) drains and interior primary drains. Irrespective of the roof slope, all bays are susceptible. Susceptible bays must be checked to preclude ponding instability and confirmed to have adequate strength with the ponding load.

The limits of 1/4 in. per ft and 1 in. per ft are based upon a maximum deflection to span ratio of 1/240. It can be shown that for secondary members parallel to the free draining edge, the minimum rise (β in inches) for a run of 1 ft for no impounded water/free drainage is

$$\beta = (L_s/S + \pi)/20$$

where L_s is the span of the secondary members and S is the spacing of the secondary members ($\beta = 0.76$ for $L_s = 60$ ft and $S = 5$ ft). For secondary members perpendicular to the free

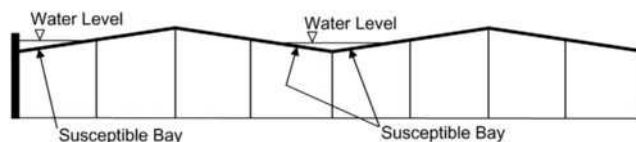


FIGURE C8.4-1 Susceptible Bays for Ponding Evaluation

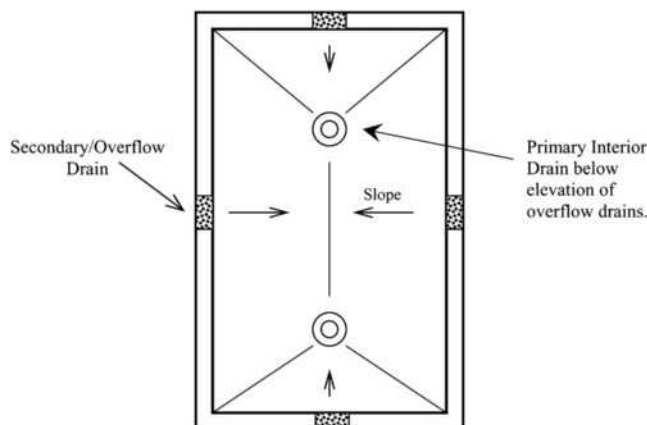


FIGURE C8.4-2 All Bays Susceptible to Ponding

draining edge, the minimum rise (β in inches) for a run of 1 ft for no impounded water/free drainage is

$$\beta = (1 + (L_p/2L_s))/10$$

where L_p is the span of the primary members ($\beta = 0.175$ for $L_s = 40$ ft and $L_p = 60$ ft).

C8.5 CONTROLLED DRAINAGE

In some areas of the country, ordinances are in effect that either limit the rate or delay the release of rainwater flow from roofs into storm drains. Controlled-flow drains are often used on such roofs. Those roofs must be capable of supporting the stormwater temporarily stored on them, similar to traditional roof drainage systems.

Examples

The following two examples illustrate the method used to establish design rain loads based on Chapter 8 of this standard.

Example 3: Determine the design rain load, R , at the secondary drainage for the roof plan shown in Fig. C8.5-1, located at a site in Birmingham, Alabama. Assume that the primary drains are blocked. The design rainfall intensity, i , based on the NOAA Precipitation Frequency Data Server (<http://hdsc.nws.noaa.gov/hdsc/pfds/index.html>) for a 100-yr, 15-min duration rainfall is 7.23 in./h (184 mm/h). The inlet of the 4-in. (102-mm) diameter secondary roof drain with an 8-in. diameter overflow dam is set 2 in. (51 mm) above the roof surface.

Flow rate, Q , for the secondary drainage 4-in. (102-mm) diameter roof drain:

$$Q = 0.0104Ai$$

$$Q = 0.0104(2,500)(7.23 \text{ in./h}) = 188 \text{ gal./min (0.0119 m}^3\text{/s)}$$

Hydraulic head, d_h :

Using Table C8.3-1 and C8.3-2 for a 4-in. (102-mm) diameter secondary roof drain with a flow rate of 188 gal./min

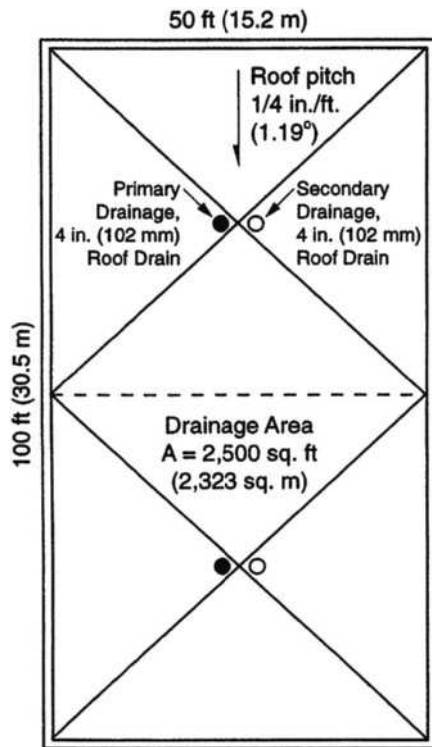


FIGURE C8.5-1 Example 3 Roof Plan

Note: Dashed line indicates the boundary between separate drainage areas.

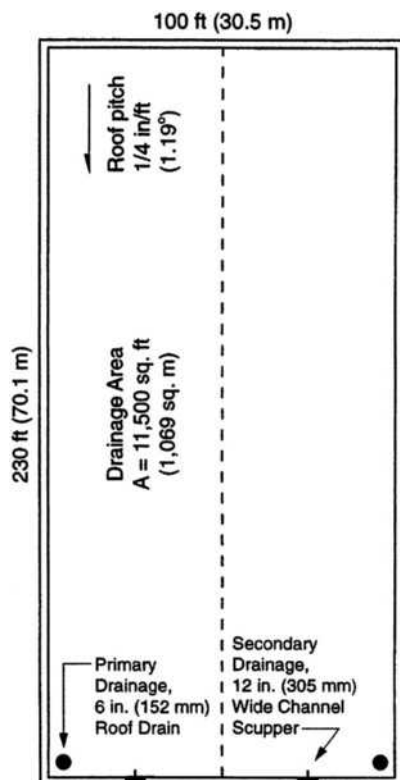


FIGURE C8.5-2 Example 4 Roof Plan

Note: Dashed line indicates the boundary between separate drainage areas.

(0.0119 m³/s), interpolate between a hydraulic head of 1.5 in. and 2 in. (38 mm and 51 mm) as follows:

$$d_h = 1.5 \text{ in.} + [188 \text{ gal./min} - 150 \text{ gal./min}] \div (200 \text{ gal./min} - 150 \text{ gal./min}) \times [2.0 \text{ in.} - 1.5 \text{ in.}] = 1.9 \text{ in.}$$

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$$d_h = 38 \text{ mm} + [0.0119 \text{ m}^3/\text{s} - 0.0095 \text{ m}^3/\text{s}] \div (0.0126 \text{ m}^3/\text{s} - 0.0095 \text{ m}^3/\text{s}) \times [51 \text{ mm} - 38 \text{ mm}] = 48.2 \text{ mm}$$

Static head $d_s = 2 \text{ in. (51 mm)}$, the water depth from drain inlet to the roof surface.

Design rain load, R , adjacent to the drains:

$$R = 5.2(d_s + d_h) \quad (8.3-1)$$

$$R = 5.2(2 + 1.9) = 20.3 \text{ psf}$$

$$R = 0.0098(d_s + d_h) \quad (8.3-1.si)$$

$$R = 0.0098(51 + 48.2) = 0.97 \text{ kN/m}^2$$

Example 4: Determine the design rain load, R , at the secondary drainage for the roof plan shown in Fig. C8.5-2, located at a site in Los Angeles, California. Assume that the primary drains are blocked. The design rainfall intensity, i , based on the NOAA Precipitation Frequency Data Server (<http://hdsc.nws.noaa.gov/hdsc/pfds/index.html>) for a 100-yr, 15-min duration rainfall is 3.28 in./h (83 mm/h). The inlet of the 12-in. (305-mm) secondary roof scuppers is set 2 in. (51 mm) above the roof surface.

Flow rate, Q , for the secondary drainage, 12-in. (305-mm) wide channel scupper:

$$Q = 0.0104Ai$$

$$Q = 0.0104(11,500)(3.28) = 392 \text{ gal./min (0.0247 m}^3/\text{s)}.$$

Hydraulic head, d_h :

Using Tables C8.3-3 and C8.3-4, by interpolation, the flow rate for a 12 in. (305 mm) wide channel scupper is twice that of a 6 in. (152 mm) wide channel scupper. Using Tables C8.3-3 and C8.3-4, the hydraulic head, d_h , for one-half the flow rate, Q , or 196 gal./min (0.0124 m³/s), through a 6 in. (152 mm) wide channel scupper is 3 in. (127 mm).

$d_h = 5 \text{ in. (127 mm)}$ for a 12-in. (305-mm) wide channel scupper with a flow rate, Q , of 392 gal./min (0.0247 m³/s).

Static head, $d_s = 2 \text{ in. (51 mm)}$, depth of water from the scupper inlet to the roof surface.

Design rain load, R , adjacent to the scuppers:

$$R = 5.2(2 + 5) = 36.4 \text{ psf}$$

In S.I.

$$R = 0.0098(51 + 127) = 1.7 \text{ kN/m}^2$$

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